Edutech Solution

AI & ML INTERNSHIP

Task 1: Understanding Dataset & Data Types

1. What is the difference between structured and unstructured data?

Structured Data

Organized in rows and columns (tables)

Easy to store and analyze

Example: Excel, SQL databases

Unstructured Data

No fixed format or structure

Harder to process

Example: Images, videos, text, social media posts

2. Explain Numerical vs Categorical data types

Numerical Data

Numbers with meaning

Used for calculations

Example: Age, salary, marks

Categorical Data

Represents categories/labels

Not used for mathematical operations

Example: Gender, City, Color

3. What are continuous and discrete variables?

Continuous Variables

Can take any value (including decimals)

Example: Height (170.5 cm), Weight

Discrete Variables

Countable values (whole numbers)

Example: Number of students, number of cars

4. How do you handle missing values in a dataset?

Common methods:

Remove rows/columns with missing data

Fill with:

Mean (average)

Median

Mode

Forward fill / backward fill

Use prediction models (advanced)

Example (Pandas):

Python

```
df.fillna(df.mean(), inplace=True)
```

5. What does the .describe() function tell you?

.describe() gives summary statistics of numerical data:

Count (number of values)

Mean (average)

Standard deviation (spread)

Minimum value

25%, 50%, 75% (quartiles)

Maximum value

Example:

Python

```
df.describe()
```

Sample Dataset (CSV)

Save this as student_data.csv

Csv

```
Name,Age,Gender,Marks,City
Kiran,21,Male,85,Hyderabad
Anu,20,Female,90,Vijayawada
Ravi,22,Male,78,Visakhapatnam
Sita,21,Female,,Hyderabad
Rahul,23,Male,88,
Priya,20,Female,92,Vijayawada
```

✅ Python Code (Task Implementation)

Python

```python
import pandas as pd


# 1. Load dataset
df = pd.read_csv("student_data.csv")


# 2. Display first rows
print("First 5 Rows:\n", df.head())


# 3. Check data types
print("\nData Types:\n", df.dtypes)


# 4. Identify Numerical vs Categorical
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
categorical_cols = df.select_dtypes(include=['object']).columns
```

```python
print("\nNumerical Columns:", numerical_cols)
print("Categorical Columns:", categorical_cols)

# 5. Dataset info
print("\nDataset Info:")
df.info()

# 6. Summary statistics
print("\nSummary Statistics:\n", df.describe())

# 7. Check missing values
print("\nMissing Values:\n", df.isnull().sum())

# 8. Fill missing values
df['Marks'].fillna(df['Marks'].mean(), inplace=True)
df['City'].fillna("Unknown", inplace=True)

print("\nAfter Handling Missing Values:\n", df)

# 9. Unique values in categorical columns
for col in categorical_cols:
    print(f"\nUnique values in {col}: {df[col].unique()}")

# 10. Shape of dataset
print("\nShape of dataset:", df.shape)
```

✅ Short README Summary (for GitHub)

You can paste this in README.md:

Task 1: Understanding Dataset & Data Types

Objective

Analyze a dataset using Pandas and understand data types.

Steps Performed

- Loaded dataset using Pandas
- Identified numerical and categorical columns
- Used .info() and .describe() for analysis
- Checked missing values
- Handled missing values using mean and fill methods
- Extracted unique values from categorical columns
- Checked dataset shape

Tools Used

- Python
- Pandas
- NumPy

Outcome

Gained basic understanding of exploratory data analysis (EDA).